

Learning goals:

- Familiarize yourself with Python and Jupyter notebooks;
- Familiarize yourself with posting on the Brightspace forum;
- Test and refresh your relevant background knowledge.

Question 1: Set up the BCI Python environment

Follow the instructions from the `bci_python_environment_setup.pdf` to set up your Python environment. All assignments will work from that environment.

Question 2: Familiarize yourself with Jupyter notebooks and MNE

Now that you have a working Python setup, open, run through the **Jupyter notebooks** and **MNE Jupyter notebooks**, and make yourself fluent in using notebooks and the basic of MNE.

Question 3: Make a Brightspace forum post about a BCI application

On the Brightspace forum, we have set up separate pages for individual weeks. To get into the habit of posting frequently on the Brightspace forum, let's make a forum post right away:

- (a) Find an interesting application of BCI and post the link to it (e.g., a news item, blog post, scientific article, podcast, video, etc.) in the forum of this week. Briefly, in about two sentences, describe the application. As the subject of your posts, use a sensible title that explains the body. Please, add your group number.
- (b) Comment on at least one of the applications of other groups with interesting additional information or a critical question.

Question 4: Do the entry self-test

Do the entry self-test to identify background gaps in your understanding that need refreshment.

Question 5: Refresh your background

Knowing about your gaps in understanding from the self-test, go back to the courses teaching these concepts or any online resource to refresh that background. Doing so will make doing the BCI course much easier, if not at all feasible.

- (a) **Linear algebra**
 - SOW-BKI124 Linear Algebra
 - [vectors](#) and [matrices](#)
 - [matrix transformations](#) and [inverse](#)
 - eigenvalue decomposition: [3Blue1Brown](#), [Khan Academy](#)
 - [University of Waterloo matrix cookbook](#).
 - [3Blue1Brown's Essence of linear algebra series](#)
 - [Khan Academy](#)
- (b) **Calculus**
 - SOW-BKI104 Calculus I
 - SOW-BKI260 Calculus II
 - [3Blue1Brown's Essence of calculus series](#)
- (c) **Probability**
 - SOW-BKI137 Probability Theory
 - SOW-BKI138 Frequentist Statistics
 - SOW-BKI203 Bayesian Statistics
 - [Steve Brunton's Probability bootcamp](#)

(d) **Machine learning**

- SOW-BKI262 From Data to Model
- SOW-BKI230A Deep Learning
- [Towards Data Science blog post](#)

(e) **Digital signal processing**

- SOW-BKI144 Introduction Digital Signal Processing
- [Stanford lecture notes](#).

Question 6: Submit to Brightspace

Submit a single, maximum one-page, document (pdf extension) describing per student in the group whether you succeeded setting up the Python environment (or if not, what went wrong), the gaps that you identified, and how you refreshed those gaps.

Please add a line on whether, where, and how you used GenAI for this assignment.